Boustead Projects E&C Pte Ltd
#07-01 Edward Boustead Centre
82 Ubi Avenue 4, Singapore 408832
Tel: +65 6748 3945 Fax: +65 6748 9250

JOB / WORK COMPLETION REPORT

PROJECT DETAILS			
Client Company Name:	Applied Materials Southest Asia Pte Ltd	Date:	16.12.2024
Project Name:	AMAT @TIC	PO/VO No.:	AGMT 10723
Project Addess:	5 Tampines Industrial Cres, Singapore 529557		
Service Completion Date:	13.12.2024	Ref No. :	BP-22207-MCI-037

SERVICE DETAILS

This is to confirm/certify that the following job/services

S/N	Quantity	PO Description
1		System Integration between BMS and Digital Twin
2		Modules Deployment Digital Twin PC Application (Sandbox)

Remarks: With the T&C being selective for the points, note some comments the below:
Approved with comments (for sandbox application) as per follows:
1. To ensure the data sending in from the BMS tallies with DataMesh
2. To ensure all the assets from the BMS/Forge is included in DataMesh e.g. Genset and P&S pumps
3. One of the feature is to have the historical graphical representation in DataMesh (to add this in)

CLIENT ACKNOWLEDGEDMENT

We acknowledged the above job/services has been fully completed to our satisfaction.

Boustead Projects Representative		Client Representative	
Date:	17 Dec 2024	Date:	19.12.2024
Name:	Tan Wee Hong	Name:	Tham Sin Pin
Signature:		Signature:	
Company Stamp	BOUSTEAD PROJECTS E&C PTE. LTD.	Company Stamp Applied Materials Southeast Asia Pte Ltd	

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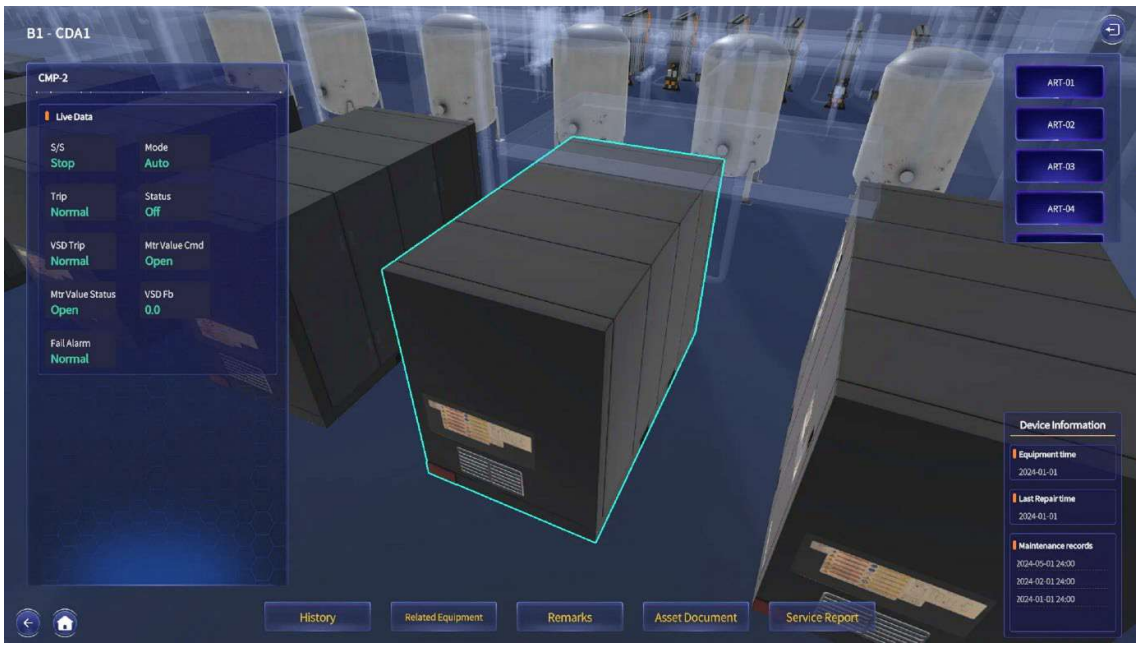
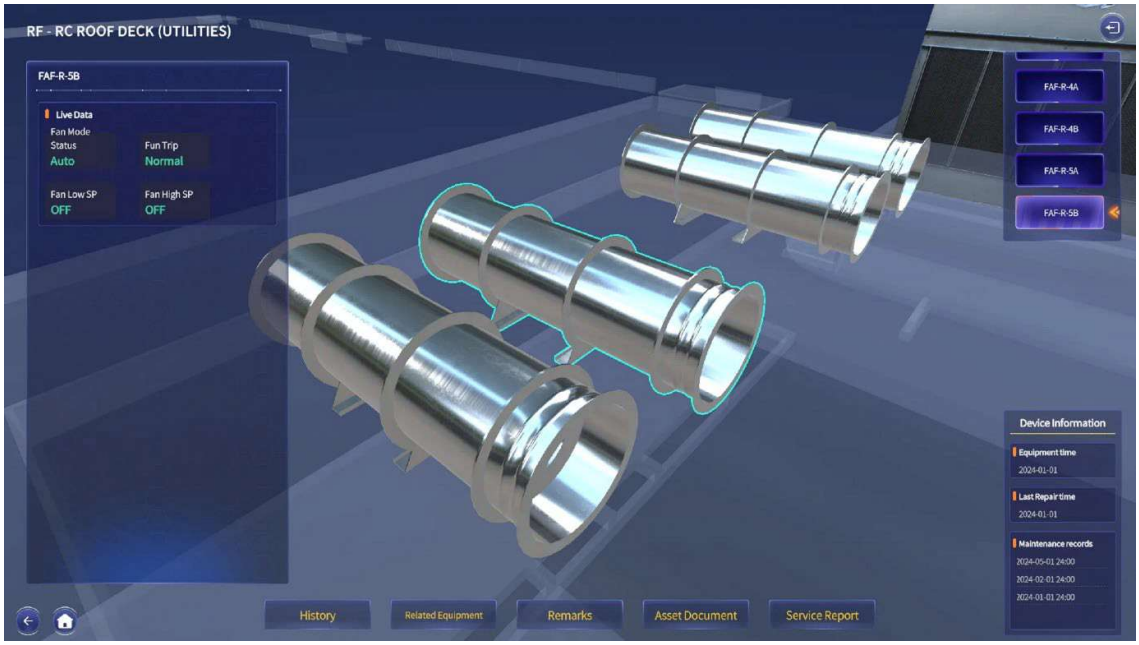
1. PROGRESS PHOTOS – Completion photos of Digital Twin PC Applications with BMS DATA Intergation (Completion Date: 13Dec2024)



Digital Twin PC Applications (Sandbox Modules) with BMS DATA



Digital Twin PC Applications (Sandbox Modules) with BMS DATA

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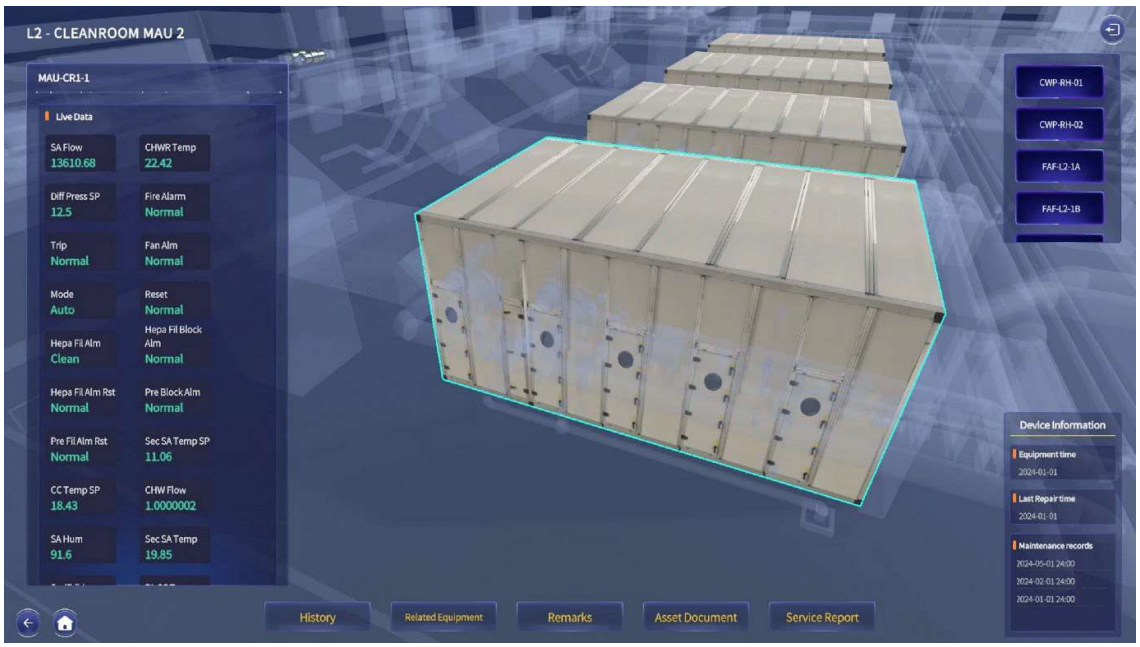

Digital Twin PC Applications (Sandbox Modules) with BMS DATA

Digital Twin PC Applications (Sandbox Modules) with BMS DATA

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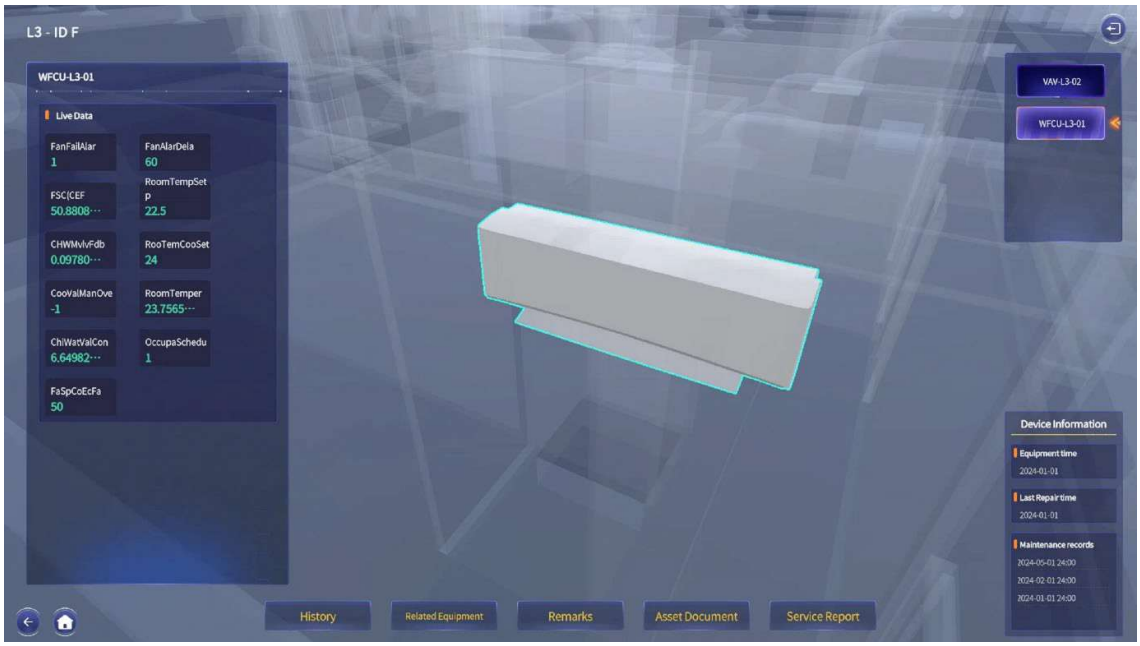
Digital Twin PC Applications (Sandbox Modules) with BMS DATA



Digital Twin PC Applications (Sandbox Modules) with BMS DATA

APPLIED MATERIALS @ TAMPINES
Digital Twin Solution I/O TEST PACK

Digital Twin PC Application (Sandbox) Graphic Display	LEVELs	Physical Set Up (Servers) with software & VM ware installation Servers Delivers on Sites with all the Software requires like OS Software, partitions and VM installation completed	Remark
Item		Note: √ As per completed in Digital Twin software.	
192.168.20.30	B1	✓	
192.168.20.40	B1	✓	

Tested by:	Steve Lee	Trade Contractor:	Lionel Quak	Witness by:	Tan Wee Hong
Signature:		Signature:		Signature:	
Company:	DataMesh Pte Ltd	Company:	Honeywell	Company:	Boustead
Date:	13Dec2024	Date:	13Dec2024	Date:	13Dec2024

APPLIED MATERIALS @ TAMPINES
Digital Twin Solution I/O TEST PACK

Digital Twin PC Application (Sandbox) Graphic Display	LEVELs	Construction of Architectural Structural Mode Based on the BIM model, coupled with data and image materials collected on-site, create and optimize architectural structural and exterior models	Remark
Item		Note: √ As per completed in Digital Twin software.	
Architecture	B1	✓	
Structure	B1	✓	
Exterior facade of each floor	B1	✓	
Architecture	L1	✓	
Structure	L1	✓	
Exterior facade of each floor	L1	✓	
Architecture	L2	✓	
Structure	L2	✓	
Exterior facade of each floor	L2	✓	
Architecture	L3	✓	
Structure	L3	✓	
Exterior facade of each floor	L3	✓	
Architecture	RF	✓	
Structure	RF	✓	
Exterior facade of each floor	RF	✓	

Tested by:	Steve Lee	Trade Contractor:	Lionel Quak	Witness by:	Tan Wee Hong
Signature:		Signature:		Signature:	
Company:	DataMesh Pte Ltd	Company:	Honeywell	Company:	Boustead
Date:	13Dec2024	Date:	13Dec2024	Date:	13Dec2024

APPLIED MATERIALS @ TAMPINES
Digital Twin Solution I/O TEST PACK

Digital Twin PC Application (Sandbox) Graphic Display	LEVELS	Creation of various professional pipeline model Following the BIM model, create and optimize various professional pipeline models.	Remark
Item		Note: √ As per completed in Digital Twin software.	
ACMV Pipeline	B1	✓	Pipeline (ductwork) model are completed. Pipework model need AMAT Sinpin advise whether to include in Digital Twin model.
ACMV Pipeline	L1	✓	
ACMV Pipeline	L2	✓	
ACMV Pipeline	L3	✓	
Fire Pipeline	B1	✓	
Fire Pipeline	L2	✓	
Rodi Pipeline	B1	✓	
EHPS pipeline	B1	✓	

Tested by:	Steve Lee	Trade Contractor:	Lionel Quak	Witness by:	Tan Wee Hong
Signature:		Signature:		Signature:	
Company:	DataMesh Pte Ltd	Company:	Honeywell	Company:	Boustead
Date:	13Dec2024	Date:	13Dec2024	Date:	13Dec2024

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APPLIED MATERIALS @ TAMPINES
Digital Twin Solution I/O TEST PACK

DataMesh

Graphic Display		Result	Remark
Name		Note: ✓ means test passed at Digital Twin software.	

Tested by:	Steve Lee	Trade Contractor:	Lionel Quah	Witness by:	[illegible]
Signature:		Signature:		Signature:	
Company:	DataMesh Pte Ltd	Company:	Honeywell	Company:	Boustead
Date:	[illegible]	Date:	[illegible]	Date:	[illegible]

APPLIED MATERIALS @ TAMPINES

Digital Twin Solution I/O TEST PACK

DataMesh

[illegible]	[illegible]	[illegible]	Remark
Item		[illegible]	

Tested by:	Steve Lee	Trade Contractor	Lionel Quek	Witness by:	Tan Wee Hong
Signature:	[illegible]	Signature:	[illegible]	Signature:	[illegible]
Company:	DataMesh Pte Ltd	Company:	Honeywell	Company:	Boustead
Date:	[illegible]	Date:	[illegible]	Date:	[illegible]

DataMesh

Digital Twin at Applied Materials TIC
Digital Twin System Attendance

Client Company : **APPLIED MATERIALS SOUTH EAST ASIA PTE LTD**

Project : **DESIGN, FABRICATION, SUPPLY, DELIVERY,INSTALLATION, TESTING, COMMISSIONING,COMPLETION, AND WARRANT OF BUILDING MANAGEMENT SYSTEM ("BMS")**
PROPOSED ERECTION OF A 3-STOREY SINGLE-USER FACTORY BUILDING WITH WAREHOUSE, ANCILLARY OFFICE AND BASEMENT CAR PARK ON LOT 03205N PT MK29
TAMPINES INDUSTRIAL CRESCENT
(PAYA LEBAR PLANNING AREA)

Date / Time : 13/Dec/2024

S/No.	Name	Job Designation	Signature
1	Lionel Quec	HUL	[signature]
2	Theo Htike Aung	BPEC	[signature]
3	B. Amirthalingam	tech	[signature]
4	Siti nurfatin	Engineer	[signature]
5	Jin Pin	Engineer	[signature]
6	Joel Neo	Engineer	[signature]
7	Wee Hong	BPEC	[signature]
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Conducted by : Steve Lee [signature] 13/Dec/2024
Data Mesh
Name / Signature / Date